

9.5 QR Decomposition



Core skills covered in this section:

- ▶ QR decomposition: decomposing any real matrix into the product of an orthogonal matrix and an upper triangular matrix.
- ▶ The connection between Gram-Schmidt orthogonalization and QR decomposition.
- ▶ Distinguishing between complete and reduced QR decomposition.
- ▶ Each column of the upper triangular matrix gives the coordinates of the original column vector in the orthogonal basis formed by the orthogonal matrix's column vectors.
- ▶ The relationship between a matrix's QR decomposition and the Cholesky decomposition of its Gram matrix.

QR decomposition and Gram-Schmidt orthogonalization have a direct and profound connection. QR decomposition's goal is to decompose a real matrix into the product of an orthogonal matrix ($\mathbf{Q}$) and an upper triangular matrix ($\mathbf{R}$). The Gram-Schmidt orthogonalization process is precisely one way to construct the orthogonal matrix $\mathbf{Q}$.

Recalling the previous section, the Gram-Schmidt algorithm takes the full-column-rank $\mathbf{A}$'s column vectors as input, and through column-by-column orthogonalization constructs a set of pairwise orthogonal, normalized vectors; this set of vectors forms the columns of $\mathbf{Q}$. At the same time, while constructing $\mathbf{Q}$, the projection coefficients of each original column vector along these orthogonal directions are computed; these coefficients are exactly the elements of matrix $\mathbf{R}$. Therefore, Gram-Schmidt not only provides a way to construct $\mathbf{Q}$, but also naturally gives the corresponding $\mathbf{R}$.

⚠ However, it's worth noting that classical Gram-Schmidt may not be numerically stable enough; therefore, in practical applications, other algorithms — such as the Householder transformation — are more commonly used to perform QR decomposition.

In this section, let's talk about QR decomposition.

Two Types of QR Decomposition

QR decomposition (QR decomposition, QR factorization) has two common forms:

- ◀ Complete (complete), $\mathbf{Q}$ is an orthogonal matrix (square matrix);
- ◀ Reduced (reduced, thin), $\mathbf{Q}$ is a semi-orthogonal matrix (same shape as the original matrix).

Let's first look at complete QR decomposition.

Figure 1 shows a schematic diagram for performing complete QR decomposition on the tall, narrow matrix $\mathbf{A}$, with the corresponding equation being:

$$\mathbf{A}_{n \times p} = \mathbf{Q}_{n \times n} \mathbf{R}_{n \times p}$$

Here, Q is an orthogonal matrix with shape $n \times n$; R and A have the same shape, $n \times p$.

⚠ Note that any real matrix can undergo QR decomposition.

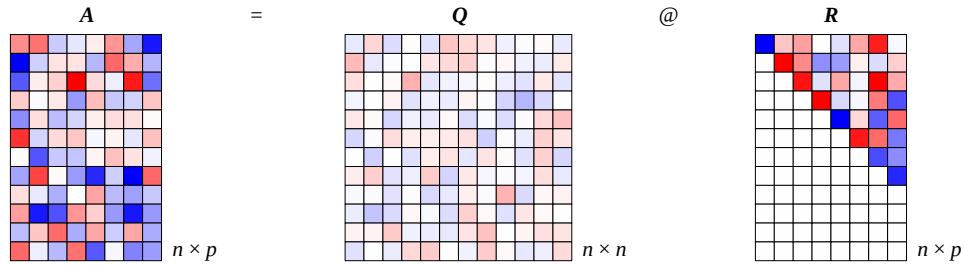


Figure 1. Schematic Diagram of Complete QR Decomposition



LA_09_05_01.ipynb performs and visualizes complete QR decomposition.

Figure 2 shows a schematic diagram of performing reduced QR decomposition on the tall, narrow matrix A , with the corresponding matrix multiplication

$$A_{n \times p} = Q_{n \times p} R_{p \times p}$$

Here, Q is a semi-orthogonal matrix.

⚠ Note that if A is a tall, narrow matrix, the Q obtained from reduced QR decomposition is a semi-orthogonal matrix; Q 's column vectors form an orthonormal basis. But if A is a flat matrix (fewer rows than columns), the situation is different. We'll talk about this at the end of this section.

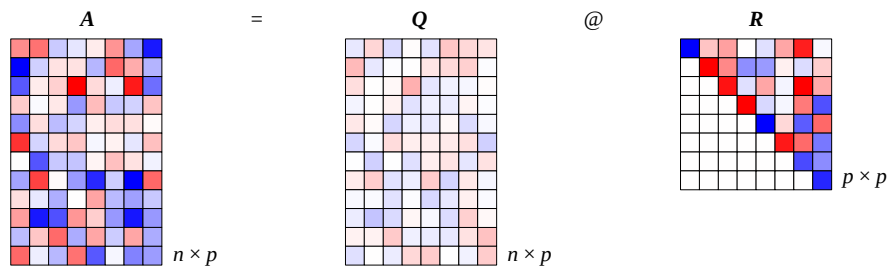


Figure 2. Schematic Diagram of Reduced QR Decomposition



LA_09_05_02.ipynb performs and visualizes reduced QR decomposition.

Q in Complete QR Decomposition

In complete QR decomposition, the square matrix Q is an orthogonal matrix, that is:

$$Q_{n \times n} Q_{n \times n}^T = Q_{n \times n}^T Q_{n \times n} = I_{n \times n}$$

Figure 3 shows the heatmap corresponding to the equation above.

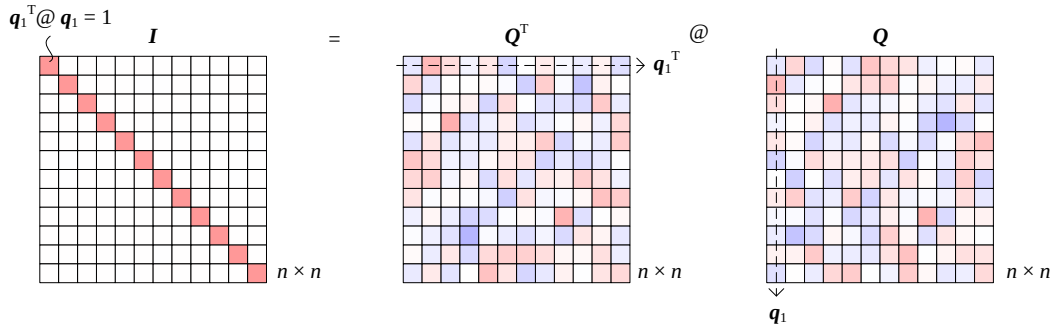


Figure 3. Heatmap of the Matrix Multiplication $Q^T @ Q = I$, Complete QR Decomposition

Figure 4 shows the Gram matrix $Q^T @ Q = I$ expanded using the first perspective of matrix multiplication. Q 's column vectors are all unit vectors and are pairwise orthogonal, i.e.,

$$Q^T Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix} = \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \cdots & q_1^T q_n \\ q_2^T q_1 & q_2^T q_2 & \cdots & q_2^T q_n \\ \vdots & \vdots & \ddots & \vdots \\ q_n^T q_1 & q_n^T q_2 & \cdots & q_n^T q_n \end{bmatrix}_{n \times n} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}_{n \times n}$$

? Please write each element of the Gram matrix above in the form of a vector inner product.

Based on the properties of orthogonal matrices introduced earlier in this book, Q 's column vectors $[q_1, q_2, \dots, q_n]$ form an orthonormal basis, spanning the vector space $\mathbb{R}^n$.

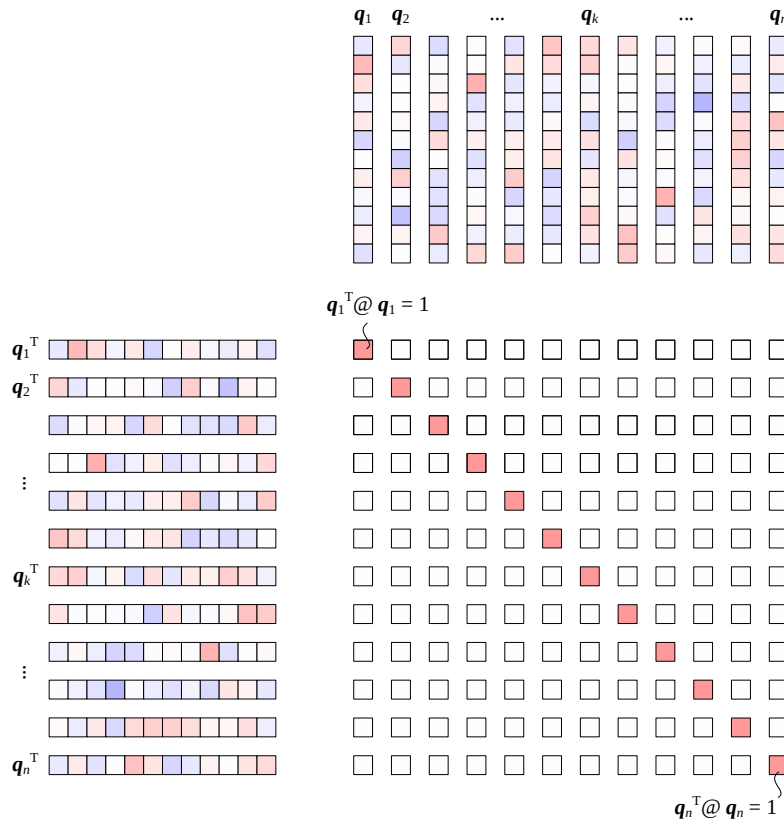


Figure 4. Matrix Multiplication $Q^T @ Q = I$, Expanded Using the First Perspective of Matrix Multiplication, Complete QR Decomposition

Figure 5 shows the heatmap of the matrix multiplication $Q @ Q^T = I$.

? Orthogonal matrix Q 's row vectors are also pairwise orthogonal unit vectors, which means Q 's row vectors are also an orthonormal basis. Please use the first perspective of matrix multiplication to expand $Q @ Q^T = I$, and explain.

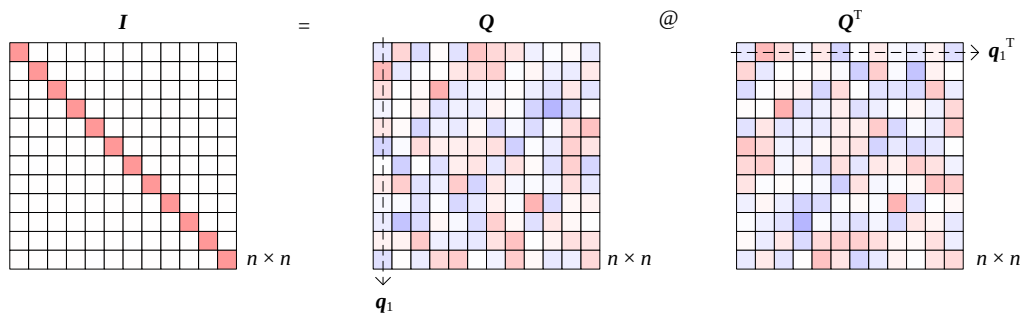


Figure 5. Matrix Multiplication $Q @ Q^T = I$ / Heatmap, Complete QR Decomposition

Using the second perspective of matrix multiplication to expand $Q @ Q^T = I$

$$QQ^T = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \end{bmatrix} \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix} = \mathbf{q}_1\mathbf{q}_1^T + \mathbf{q}_2\mathbf{q}_2^T + \cdots + \mathbf{q}_n\mathbf{q}_n^T = \mathbf{I}$$

Since $\mathbf{Q}$'s column vectors are all nonzero vectors, the $\mathbf{q}_i\mathbf{q}_i^T$ terms in the equation above are all rank-one matrices, and also symmetric matrices; for example, Figure 6 shows the rank-one matrix obtained from $\mathbf{q}_1 @ \mathbf{q}_1^T$.

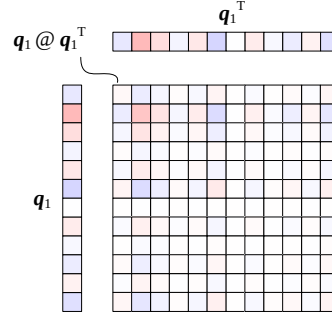


Figure 6. Matrix Multiplication $\mathbf{q}_1 @ \mathbf{q}_1^T$ Heatmap, Complete QR Decomposition

This represents n rank-one matrices superimposed to give the $n \times n$ identity matrix, specifically as Figure 7 shows. In the figure, each $\mathbf{q}_i\mathbf{q}_i^T$ is a single-direction projection matrix; together they represent orthogonal directional contributions.

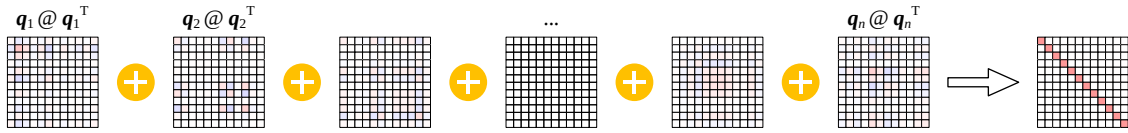


Figure 7. n Rank-One Matrices Superimposed to Give the Identity Matrix, Complete QR Decomposition

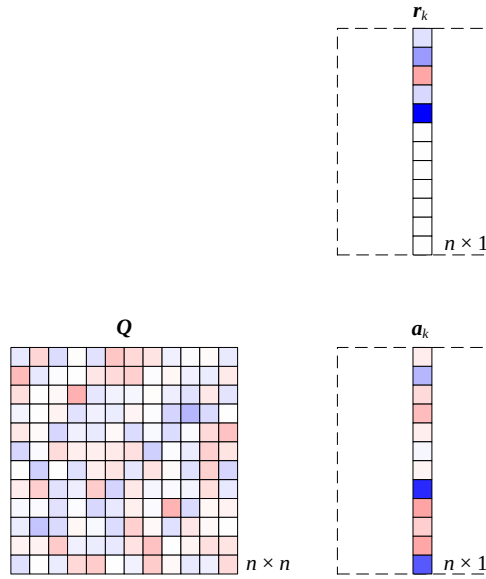
R's Column Vectors Are Coordinates

Using the third perspective of matrix multiplication — linear combination of column vectors — to analyze $\mathbf{A} = \mathbf{QR}$

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{bmatrix} = \mathbf{QR} = \mathbf{Q} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \cdots & \mathbf{r}_p \end{bmatrix} = \begin{bmatrix} \mathbf{Q}\mathbf{r}_1 & \mathbf{Q}\mathbf{r}_2 & \cdots & \mathbf{Q}\mathbf{r}_p \end{bmatrix}$$

As Figure 8 shows, expanding $\mathbf{Q}$ as $[\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n]$, $\mathbf{A}$'s column vectors $\mathbf{a}_k$ can be written as a linear combination of $\mathbf{Q}$'s column vectors, i.e.,

$$\mathbf{a}_k = \mathbf{Q}\mathbf{r}_k = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \end{bmatrix} \begin{bmatrix} r_{1,k} \\ r_{2,k} \\ \vdots \\ r_{n,k} \end{bmatrix} = r_{1,k}\mathbf{q}_1 + r_{2,k}\mathbf{q}_2 + \cdots + r_{n,k}\mathbf{q}_n$$

Figure 8. Computing $\mathbf{a}_k$

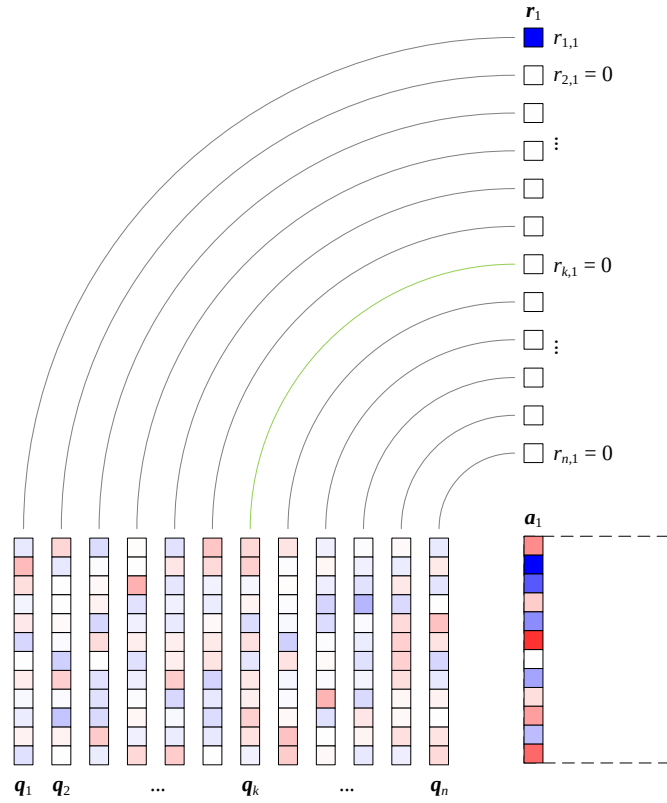
For example, the first column vector $\mathbf{a}_1$ can be obtained through the following equation:

$$\mathbf{a}_1 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_n] \begin{bmatrix} r_{1,1} \\ r_{2,1} \\ \vdots \\ r_{n,1} \end{bmatrix} = \underset{=0}{r_{1,1}\mathbf{q}_1} + \underset{=0}{r_{2,1}\mathbf{q}_2} + \cdots + \underset{=0}{r_{n,1}\mathbf{q}_n} = r_{1,1}\mathbf{q}_1$$

As Figure 9 shows, the equation above is equivalent to saying that $\mathbf{a}_1$'s coordinates in the space spanned by the orthonormal basis $[\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n]$ are $(r_{1,1}, r_{2,1}, \dots, r_{n,1})$, that is, $(r_{1,1}, 0, \dots, 0)$.

That is to say, $\mathbf{a}_1$ and $\mathbf{q}_1$ are parallel, pointing in the same or opposite direction.

This is consistent with the first step of Gram-Schmidt orthogonalization introduced in the previous section.

Figure 9. The Matrix Multiplication Corresponding to $\mathbf{A}$'s First Column, $\mathbf{a}_1 = \mathbf{Q} \mathbf{r}_1$

$\mathbf{q}_1$ is a unit vector, that is:

$$r_{1,1} = \pm \|\mathbf{a}_1\|$$

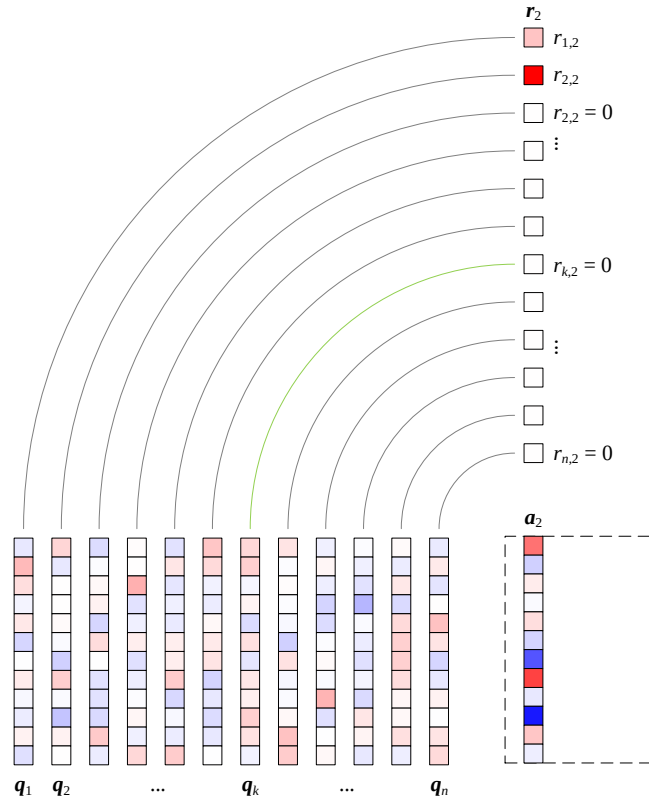
This already shows that the QR decomposition result is not unique. However, if $\mathbf{A}$ has full column rank, and $\mathbf{R}$'s diagonal elements are positive real numbers, the QR decomposition is unique.

Similarly, $\mathbf{A}$'s second column vector $\mathbf{a}_2$ is written as:

$$\mathbf{a}_2 = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_n] \begin{bmatrix} r_{1,2} \\ r_{2,2} \\ \vdots \\ r_{n,2} \end{bmatrix} = r_{1,2} \mathbf{q}_1 + r_{2,2} \mathbf{q}_2 + r_{3,2} \mathbf{q}_3 + \cdots + r_{n,2} \mathbf{q}_n = r_{1,2} \mathbf{q}_1 + r_{2,2} \mathbf{q}_2$$

$\underbrace{\hspace{1.5cm}}_{=0} \quad \underbrace{\hspace{1.5cm}}_{=0}$

As Figure 10 shows, $\mathbf{a}_2$'s coordinates in the space spanned by the orthonormal basis $[\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n]$ are $(r_{1,2}, r_{2,2}, r_{3,2}, \dots, r_{n,2})$, that is, $(r_{1,2}, r_{2,2}, 0, \dots, 0)$.

Figure 10. The Matrix Multiplication Corresponding to A 's Second Column, $\mathbf{a}_2 = Q @ \mathbf{r}_2$

Reduced QR Decomposition

Figure 1 corresponds to a complete QR decomposition that can be further simplified: cutting R horizontally so that the top sub-block is a square matrix and the bottom sub-block is the zero matrix O .

As Figure 11 shows, this can be written as block matrix multiplication:

$$A_{n \times p} = \begin{bmatrix} Q_{n \times p} & Q_{n \times (n-p)} \end{bmatrix} \begin{bmatrix} R_{p \times p} \\ O_{(n-p) \times p} \end{bmatrix} = Q_{n \times p} R_{p \times p} + Q_{n \times (n-p)} O_{(n-p) \times p} = Q_{n \times p} R_{p \times p}$$

Here, $Q_{n \times p}$ and $A_{n \times p}$ have the same matrix shape.

$Q_{n \times p}$ is a semi-orthogonal matrix. If matrix A is a tall, narrow matrix, $Q_{n \times p}$'s column vectors are unit vectors and pairwise orthogonal; that is to say, $Q_{n \times p}$'s column vectors form an orthonormal basis.

$R_{p \times p}$ is an upper triangular square matrix.

⚠ Note that in the block matrix above, the zero matrix O has shape $(n - p) \times p$, with all elements equal to 0.

Figure 11 tells us that when partitioning into blocks, in order to perform matrix multiplication, the shapes of the sub-matrices must also match.

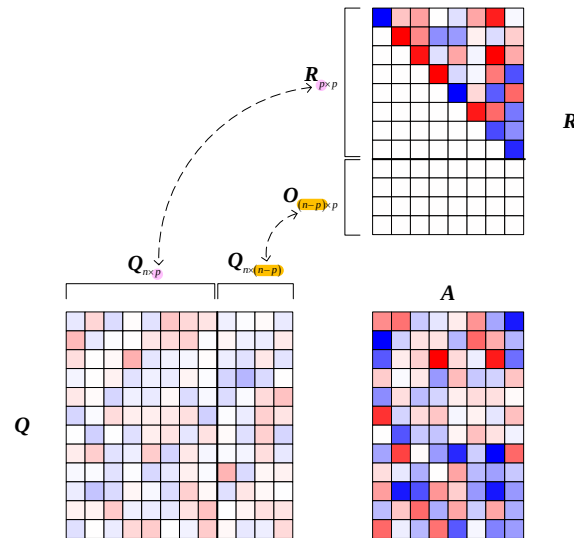


Figure 11. Block Matrix Multiplication of Complete QR Decomposition

The result of the block matrix multiplication is the sum of two matrices with the same shape; the second matrix is an all-zero matrix O , with the same shape as A . Figure 12 shows the result of these two matrix multiplications.



We especially recommend taking this opportunity to review the content on block matrix multiplication in Chapter 3 of this book.

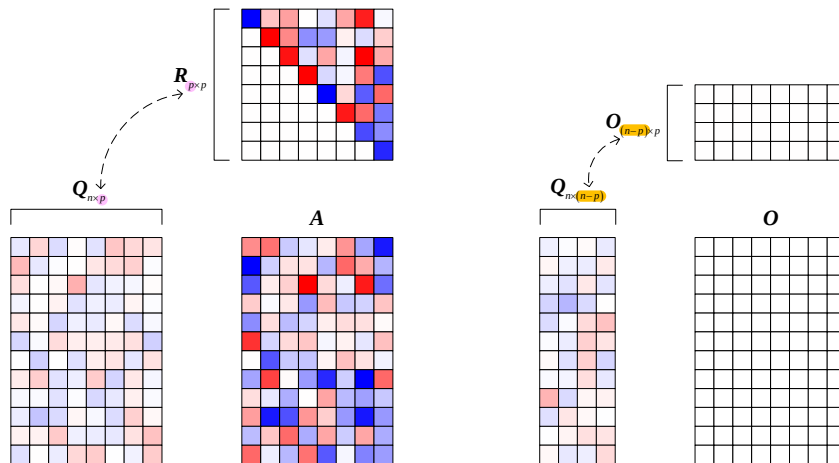


Figure 12. Sum of the Two Matrix Multiplication Results

Figure 13 shows the simplification process of QR decomposition from complete to reduced form.

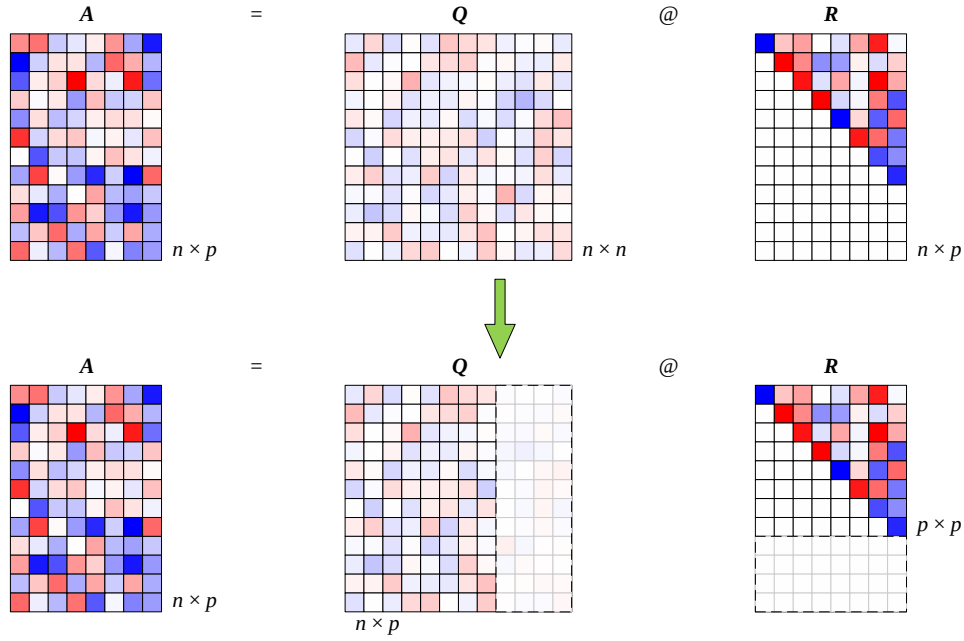


Figure 13. Simplification Process of QR Decomposition from Complete to Reduced Form

As Figure 14 shows, the spaces spanned by $Q_{n \times p}$'s column vectors and $Q_{n \times (n-p)}$'s column vectors, $\text{span}(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_p)$, and $\text{span}(\mathbf{q}_{p+1}, \mathbf{q}_{p+2}, \dots, \mathbf{q}_n)$, are mutually orthogonal complements.

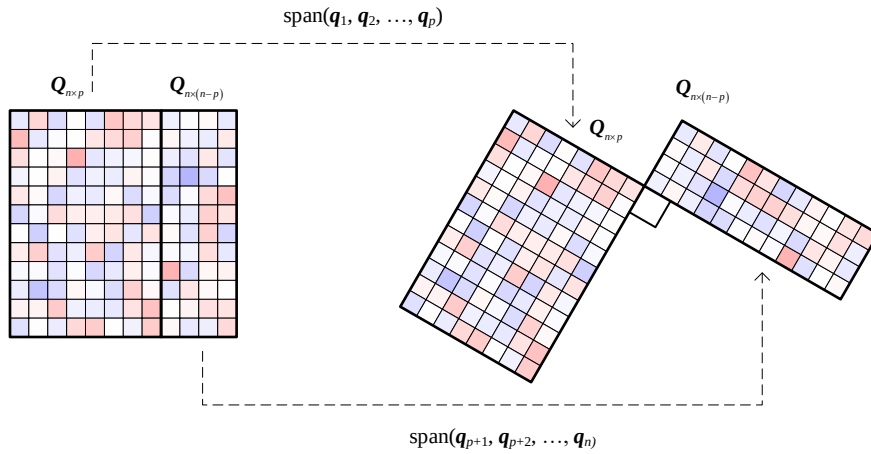


Figure 14. Vector Spaces That Are Mutually Orthogonal Complements

Geometric Perspective

From a geometric point of view, as Figure 15 shows, reduced QR decomposition completes the orthogonalization of matrix A .

A 's column vectors $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$ are quite possibly not pairwise orthogonal; the $[\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_p]$ obtained through reduced QR decomposition are pairwise orthogonal, and each vector is a unit vector.

$[\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_p]$ form an orthonormal basis.

The important characteristic of the orthonormal basis $[q_1, q_2, \dots, q_p]$ is that q_1 is parallel to a_1 .

In particular, if A has full column rank, then $\text{span}(a_1, a_2, \dots, a_p) = \text{span}(q_1, q_2, \dots, q_p)$.

Of course, the method for orthogonalizing matrix A is not unique. As mentioned earlier in this book, spectral decomposition (used for real symmetric square matrices), singular value decomposition, and others can also obtain orthogonal matrices.

⚠ Note that different orthogonalization methods yield different orthonormal bases. Later in this book, we'll introduce other orthogonalization methods; please pay attention to the differences in results and their applicable scenarios.

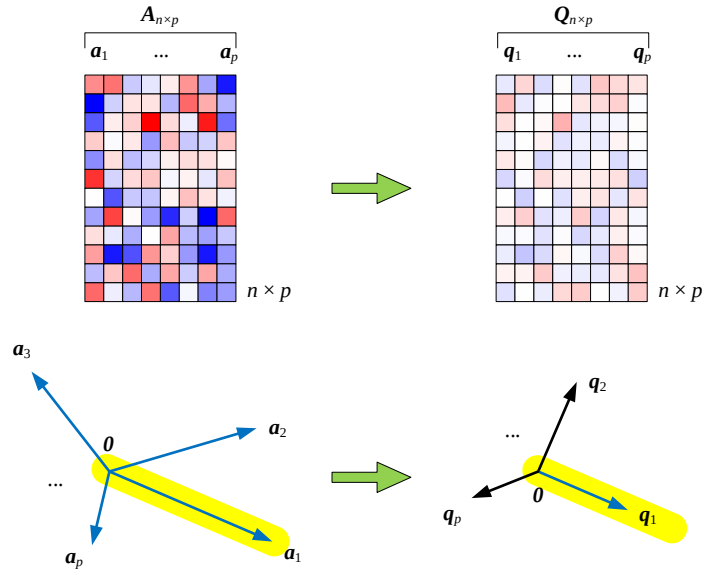


Figure 15. The Geometric Meaning Behind Reduced QR Decomposition

Semi-Orthogonal Matrix

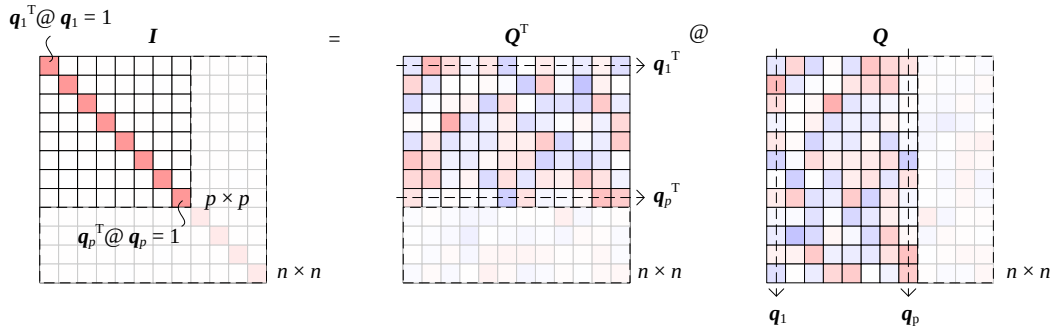
Although, within the block form above, the matrix $Q_{n \times p}$ is not a square matrix, its column vectors are still pairwise orthogonal, because,

$$(Q_{n \times p})^T Q_{n \times p} = I_{p \times p}$$

$Q_{n \times p}$ is a semi-orthogonal matrix.

⚠ Note that $Q_{n \times p}$ is no longer an orthogonal matrix. The premise of an orthogonal matrix is that the matrix is square.

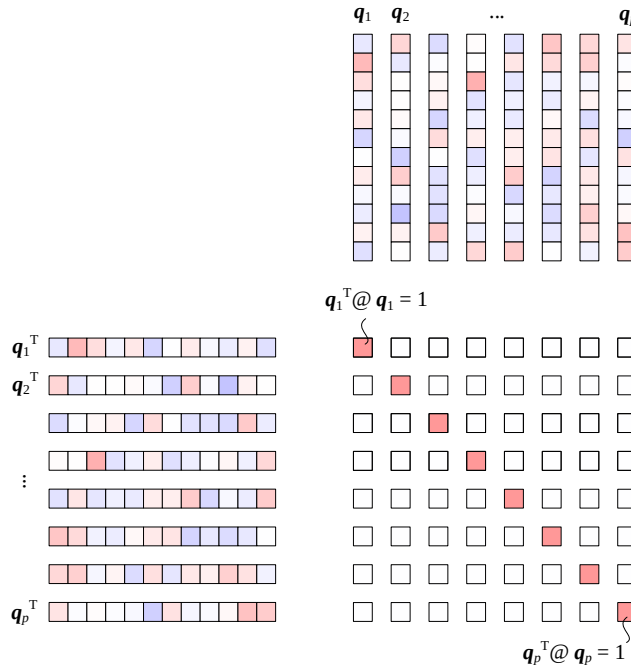
Figure 16 shows the heatmap corresponding to the $Q_{n \times p}^T Q_{n \times p}$ operation.


 Figure 16. Semi-Orthogonal Matrix $Q^T @ Q = I$

Expanding $Q_{n \times p}$ as $[q_1, q_2, \dots, q_p]$, and substituting into the equation above gives:

$$Q_{n \times p}^T Q_{n \times p} = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_p^T \end{bmatrix} \begin{bmatrix} q_1 & q_2 & \dots & q_p \end{bmatrix} = \begin{bmatrix} q_1^T q_1 & q_1^T q_2 & \dots & q_1^T q_p \\ q_2^T q_1 & q_2^T q_2 & \dots & q_2^T q_p \\ \vdots & \vdots & \ddots & \vdots \\ q_p^T q_1 & q_p^T q_2 & \dots & q_p^T q_p \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}_{p \times p}$$

Figure 17 shows how each element of $I_{p \times p}$ is obtained.


 Figure 17. Semi-Orthogonal Matrix $Q^T @ Q = I$, Expanded Using the First Perspective of Matrix Multiplication

Next, let's also use block matrix multiplication to derive how, from $Q_{n \times n}^T Q_{n \times n}$, we get $Q_{n \times p}^T Q_{n \times p}$.

As Figure 18 shows, writing the orthogonal matrix $Q_{n \times n}$ as $[Q_{n \times p}, Q_{n \times (n-p)}]$, we can expand $Q_{n \times n}^T Q_{n \times n}$ as

$$\mathbf{Q}_{n \times n}^T \mathbf{Q}_{n \times n} = \begin{bmatrix} \mathbf{Q}_{n \times p}^T \\ \mathbf{Q}_{n \times (n-p)}^T \end{bmatrix} \begin{bmatrix} \mathbf{Q}_{n \times p} & \mathbf{Q}_{n \times (n-p)} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}_{n \times p}^T \mathbf{Q}_{n \times p} & \mathbf{Q}_{n \times p}^T \mathbf{Q}_{n \times (n-p)} \\ \mathbf{Q}_{n \times (n-p)}^T \mathbf{Q}_{n \times p} & \mathbf{Q}_{n \times (n-p)}^T \mathbf{Q}_{n \times (n-p)} \end{bmatrix} = \begin{bmatrix} \mathbf{I}_{p \times p} & \mathbf{O}_{p \times (n-p)} \\ \mathbf{O}_{(n-p) \times p} & \mathbf{I}_{(n-p) \times (n-p)} \end{bmatrix}$$

Looking at this in block form, $\mathbf{Q}_{n \times p}$ is a semi-orthogonal matrix, so its Gram matrix $\mathbf{Q}_{n \times p}^T \mathbf{Q}_{n \times p}$ is the identity matrix $\mathbf{I}_{p \times p}$.

$\mathbf{Q}_{n \times (n-p)}$ is also a semi-orthogonal matrix; its Gram matrix $\mathbf{Q}_{n \times (n-p)}^T \mathbf{Q}_{n \times (n-p)}$ is the identity matrix $\mathbf{I}_{(n-p) \times (n-p)}$.

Since $\mathbf{Q}_{n \times p}$ and $\mathbf{Q}_{n \times (n-p)}$ are orthogonal to each other, $\mathbf{Q}_{n \times p}^T \mathbf{Q}_{n \times (n-p)}$ and $\mathbf{Q}_{n \times (n-p)}^T \mathbf{Q}_{n \times p}$ are both zero matrices.

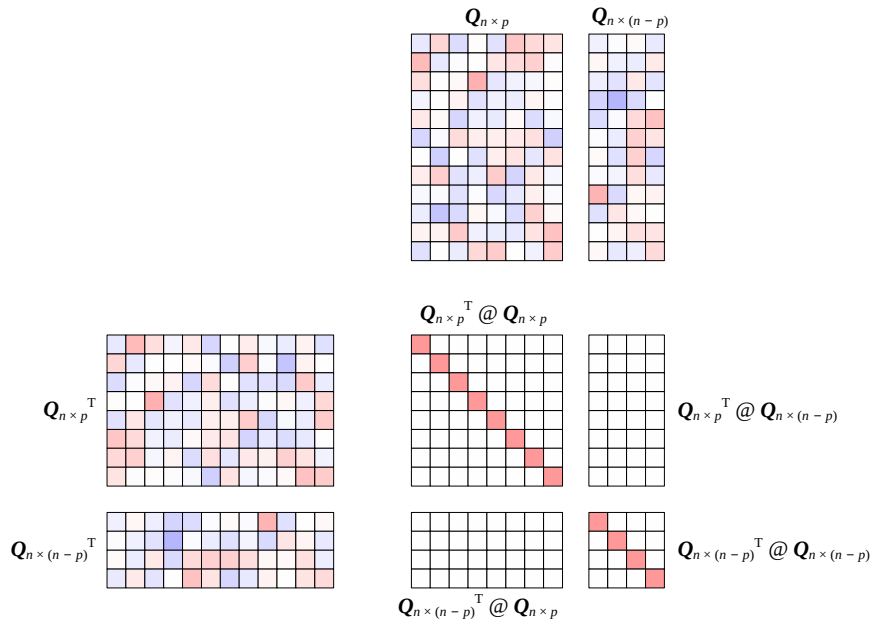


Figure 18. From $\mathbf{Q}_{n \times n}^T \mathbf{Q}_{n \times n}$, Obtaining $\mathbf{Q}_{n \times p}^T \mathbf{Q}_{n \times p}$

Figure 19 shows the heatmap corresponding to the $\mathbf{Q}_{n \times p} \mathbf{Q}_{n \times p}^T$ operation; clearly the result is not the identity matrix!

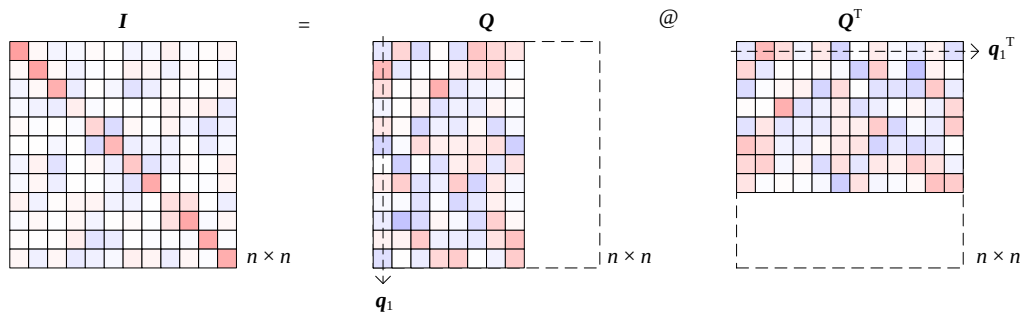


Figure 19. Semi-Orthogonal Matrix $\mathbf{Q} @ \mathbf{Q}^T = \mathbf{I}$

As Figure 20 shows, this is because $n - p$ square matrices are missing from the superposition, so the result is not the identity matrix, that is,

$$Q_{n \times p} Q_{n \times p}^T = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_p \end{bmatrix} \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_p^T \end{bmatrix} = \mathbf{q}_1 \mathbf{q}_1^T + \mathbf{q}_2 \mathbf{q}_2^T + \cdots + \mathbf{q}_p \mathbf{q}_p^T = I - (\mathbf{q}_{p+1} \mathbf{q}_{p+1}^T + \cdots + \mathbf{q}_n \mathbf{q}_n^T)$$

Here, $\mathbf{q}_1 \mathbf{q}_1^T + \mathbf{q}_2 \mathbf{q}_2^T + \cdots + \mathbf{q}_p \mathbf{q}_p^T$ is the projection matrix for orthogonal projection onto $\text{span}(\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_p)$.

$\mathbf{q}_{p+1} \mathbf{q}_{p+1}^T + \cdots + \mathbf{q}_n \mathbf{q}_n^T$ is the projection matrix for orthogonal projection onto $\text{span}(\mathbf{q}_{p+1}, \mathbf{q}_2, \dots, \mathbf{q}_n)$.

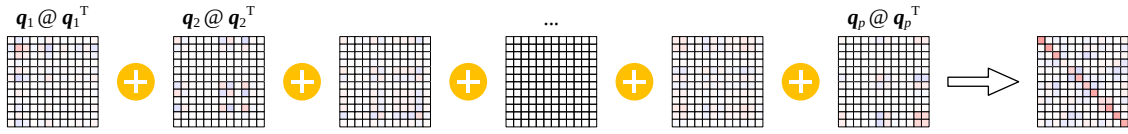


Figure 20. Square Matrix Obtained by Superimposing p Rank-One Matrices, Reduced QR Decomposition

Gram Matrix

Let's look at matrix $\mathbf{A}$'s Gram matrix $\mathbf{A}^T \mathbf{A}$, specifically as Figure 21 shows.

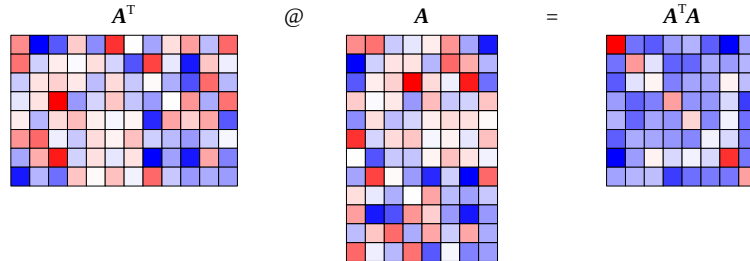


Figure 21. Computing Matrix $\mathbf{A}$'s Gram Matrix $\mathbf{A}^T \mathbf{A}$

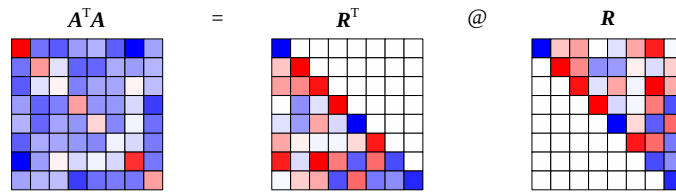
Expanding this using reduced QR decomposition gives $\mathbf{A}^T \mathbf{A}$

$$\begin{aligned} \mathbf{A}^T \mathbf{A} &= (\mathbf{Q}_{n \times p} \mathbf{R}_{p \times p})^T (\mathbf{Q}_{n \times p} \mathbf{R}_{p \times p}) \\ &= \mathbf{R}_{p \times p}^T \underbrace{\mathbf{Q}_{n \times p}^T \mathbf{Q}_{n \times p}}_I \mathbf{R}_{p \times p} \\ &= \mathbf{R}_{p \times p}^T \mathbf{R}_{p \times p} \end{aligned}$$

As Figure 22 shows, matrix $\mathbf{A}^T \mathbf{A}$ is decomposed as the product of the transpose of an upper triangular matrix ($\mathbf{R}^T$) and an upper triangular matrix ($\mathbf{R}$).



This is the Cholesky decomposition to be covered later in this book.

Figure 22. Decomposition of $A^T A$ as $R^T @ R$

Flat Matrix

Figure 23 shows the QR decomposition of a flat matrix. The flat matrix A clearly does not have full column rank.

? Please write out on your own the matrix multiplication corresponding to Figure 23.

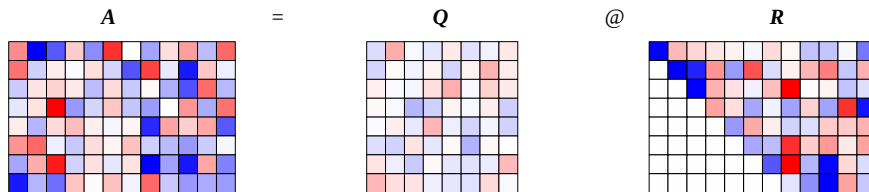
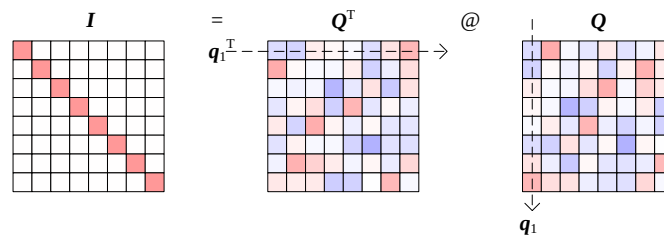
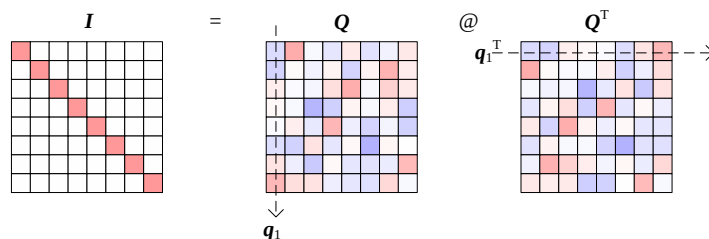


Figure 23. QR Decomposition of a Flat Matrix

As Figure 24 and Figure 25 show, the resulting Q is an orthogonal matrix.

? Please analyze Figure 24 and Figure 25 on your own.

Figure 24. $Q^T @ Q = I$, Flat MatrixFigure 25. $Q @ Q^T = I$, Flat Matrix



Please use tools such as DeepSeek/ChatGPT to complete the following exercises in this section.

Q1. Write Python code to compute the complete QR decomposition of the following matrix A .

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Q2. Learn about how to use the Householder algorithm to perform QR decomposition.